

RAZGRAD, Bulgaria

WMO: 155490

Lat: 43.567N

Lon: 26.500E

Elev: 347

StdP: 97.23

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.8	-14.2	1.1	-11.3	-11.7	1.4	-8.8	16.4	-0.2	15.7	-0.4	3.5	50	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	32.5	22.8	30.8	21.9	29.3	20.9	24.4	30.5	23.0	28.8	21.8	27.5	2.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.0	28.3	21.1	16.4	26.6	19.9	15.2	25.2	75.9	30.5	70.0	28.9	65.5	27.7	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	9.1	7.4	-14.5	35.2	3.2	2.2	-16.8	36.8	-18.7	38.1	-20.4	39.4	-22.7	41.0		
(5)			-14.8	25.5	3.1	2.7	-17.1	27.4	-18.9	29.0	-20.6	30.5	-22.9	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	0.0	2.3	6.5	11.5	16.4	20.2	22.6	22.8	18.1	12.3	7.2	1.8		
	DBStd	9.13	5.79	5.99	5.11	4.15	3.54	3.29	2.97	3.11	3.95	4.83	5.32	5.46		
	HDD10.0	1105	311	221	132	29	2	0	0	0	1	33	116	259		
	HDD18.3	2791	568	449	367	208	80	18	3	4	52	192	335	514		
	CDD10.0	1781	2	5	23	74	200	307	391	396	242	104	32	4		
	CDD18.3	424	0	0	0	3	19	75	136	141	44	6	1	0		
	CDH23.3	3456	0	0	1	21	142	578	1164	1203	311	34	1	0		
	CDH26.7	1120	0	0	0	1	15	158	422	438	81	4	0	0		
(14) Wind	WSAvg	2.9	3.0	3.3	3.6	3.3	2.8	2.5	2.6	2.6	2.8	2.7	2.9	3.0		
(15) Precipitation	PrecAvg	651	42	36	46	50	67	71	65	51	70	53	51	50		
	PrecMax	1064	94	102	92	130	175	173	198	165	219	103	106	140		
	PrecMin	379	8	3	2	4	9	10	8	1	1	1	1	2		
	PrecStd	159	23	20	24	24	41	36	54	39	54	29	28	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	18.3	21.8	25.9	28.5	32.3	35.9	34.8	31.4	26.8	22.1	15.9		
		MCWB	9.9	11.5	13.3	17.4	19.1	23.0	25.5	23.2	21.4	18.0	15.5	10.8		
	2%	DB	11.9	15.1	19.0	22.9	26.4	30.2	32.6	32.7	29.0	23.7	18.9	13.4		
		MCWB	8.3	9.8	12.0	15.5	18.3	22.3	23.0	22.7	19.5	16.5	14.1	9.9		
	5%	DB	9.9	12.6	16.7	20.3	24.9	28.4	30.8	30.8	26.8	21.4	16.6	11.2		
		MCWB	7.2	8.5	10.5	13.8	17.5	20.9	21.9	21.7	18.6	15.5	12.6	8.3		
	10%	DB	7.9	10.3	14.3	18.2	23.2	26.7	28.9	29.1	24.7	19.1	14.4	9.2		
		MCWB	5.7	7.1	9.2	12.7	16.7	19.9	21.2	20.8	17.6	14.3	11.4	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	12.1	14.4	18.2	20.7	25.8	28.5	25.6	22.4	18.7	16.5	12.2		
		MCDB	13.3	16.1	20.1	24.7	26.1	30.6	33.6	32.2	30.1	25.3	19.7	14.7		
	2%	WB	8.7	10.2	12.5	16.1	19.4	23.2	24.8	24.0	20.6	17.2	14.5	10.3		
		MCDB	11.4	14.4	17.8	21.8	24.8	28.5	30.4	31.0	27.1	22.6	18.2	12.8		
	5%	WB	7.4	8.7	11.0	14.5	18.2	21.7	23.1	22.8	19.3	15.9	13.0	8.6		
		MCDB	9.7	12.2	16.0	19.3	23.5	26.9	28.6	28.8	24.7	20.4	16.0	10.8		
	10%	WB	5.8	7.2	9.6	13.1	17.1	20.5	21.8	21.6	18.2	14.8	11.7	7.0		
		MCDB	7.7	10.1	13.9	17.3	22.1	25.5	27.2	27.4	23.3	18.3	14.0	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	7.0	8.1	9.0	9.7	9.7	10.1	10.1	9.3	8.0	6.6	5.8		
		MCDBR	8.7	10.5	11.5	11.9	11.9	11.7	12.6	12.1	12.2	11.1	9.5	8.4		
	5% WB	MCWBR	5.7	6.4	6.5	6.6	6.3	5.7	5.7	5.2	5.1	5.2	5.3	5.6		
		MCWBR	8.3	9.9	10.8	10.8	11.2	10.9	11.5	11.3	11.3	10.5	8.7	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.361	0.397	0.440	0.416	0.423	0.431	0.443	0.413	0.381	0.354	0.333		
	taud		2.316	2.251	2.203	2.091	2.232	2.244	2.228	2.194	2.248	2.323	2.351	2.330		
	Ebn at Noon		781	817	834	825	858	851	838	813	807	785	747	739		
	Edh at Noon		84	106	126	153	137	136	136	137	120	98	81	76		
(44) All-Sky Solar Radiation	RadAvg		1.49	2.25	3.37	4.65	5.79	6.35	6.48	5.82	4.23	2.74	1.62	1.24		
	RadStd		0.23	0.31	0.31	0.52	0.46	0.48	0.46	0.31	0.34	0.28	0.22	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.50	N/A	N/A	N/A	+1.37	+1.71	N/A	-232	N/A	N/A	
(47) Regional (45 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-109	+78	+49	

Nomenclature: See separate page